

MA2006B - Modern Algebras for Security and Cryptography

Lecture Notes

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Contents

1	Group Theory	3
1.1	Introduction	3
1.2	Definition and Examples of Groups	3
1.2.1	Theory	3
1.2.2	Solved Problems	4
1.2.3	Supplementary Problems	5
1.3	Subgroups	5
1.3.1	Theory	5
1.3.2	Solved Problems	6
1.3.3	Supplementary Problems	7
1.4	Homomorphisms	7
1.4.1	Theory	7
1.4.2	Solved Problems	8
1.4.3	Supplementary Problems	9
1.5	Automorphisms	9
1.5.1	Theory	9
1.5.2	Solved Problems	10
1.5.3	Supplementary Problems	11
1.6	Cayley's Theorem	11
1.6.1	Theory	11
1.6.2	Solved Problems	12
1.6.3	Supplementary Problems	13
1.7	Permutation Groups	13
1.7.1	Theory	13
1.7.2	Solved Problems	14
1.7.3	Supplementary Problems	15
2	Ring Theory	15
2.1	Introduction	15
2.2	Definition and Examples of Rings	16
2.2.1	Theory	16
2.2.2	Solved Problems	17

2.2.3	Supplementary Problems	18
2.3	Ideals and Quotient Rings	18
2.3.1	Theory	18
2.3.2	Solved Problems	19
2.3.3	Supplementary Problems	20
2.4	Euclidean Rings	20
2.4.1	Theory	20
2.4.2	Solved Problems	21
2.4.3	Supplementary Problems	22
2.5	Polynomial Rings	22
2.5.1	Theory	22
2.5.2	Solved Problems	23
2.5.3	Supplementary Problems	24

1 Group Theory

1.1 Introduction

Group theory is the study of algebraic structures known as groups. The concept of a group is central to abstract algebra and has fundamental applications in cryptography, particularly in public-key cryptosystems like RSA and Elliptic Curve Cryptography (ECC). Understanding groups, subgroups, and mappings between them (homomorphisms) is essential for grasping the security mechanisms of modern encryption protocols.

1.2 Definition and Examples of Groups

1.2.1 Theory

Definition 1.2.1.1 (Group): A **group** is a set G equipped with a binary operation $\cdot : G \times G \rightarrow G$ satisfying the following axioms:

1. **Closure:** For all $a, b \in G$, the result $a \cdot b$ is also in G .
2. **Associativity:** For all $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Identity Element:** There exists an element $e \in G$ such that for every element $a \in G$, $e \cdot a = a \cdot e = a$.
4. **Inverse Element:** For each $a \in G$, there exists an element $b \in G$, commonly denoted a^{-1} , such that $a \cdot b = b \cdot a = e$.

If the operation is also commutative (i.e., $a \cdot b = b \cdot a$ for all $a, b \in G$), the group is called an **abelian group**.

Example 1.2.1.1: The set of integers $\mathbb{Z}$ under addition is an infinite abelian group. The identity is 0, and the inverse of a is $-a$.

Example 1.2.1.2: The set $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ under addition modulo n is a finite abelian group of order n .

Example 1.2.1.3: Cryptography: Groups are the foundation of many cryptographic primitives.

- **Diffie-Hellman Key Exchange** relies on the difficulty of the Discrete Logarithm Problem in a cyclic group.
- **RSA** uses the multiplicative group of integers modulo n , denoted $\mathbb{Z}_n^*$.

1.2.2 Solved Problems

Solved Problem 1.2.2.1: Modular Addition

In $\mathbb{Z}_{26}$ (often used for classical ciphers), compute $15 + 18 \bmod 26$.

Solution:

$$15 + 18 = 33$$

Dividing by 26: $33 = 1 \cdot 26 + 7$.

$$33 \equiv 7 \pmod{26}$$

■

Solved Problem 1.2.2.2: Verify if the set of non-zero real numbers $\mathbb{R}^*$ under multiplication forms a group.

Solution: Yes, $(\mathbb{R}^*, \cdot)$ is a group.

- **Closure:** The product of two non-zero real numbers is non-zero ($a \neq 0, b \neq 0 \Rightarrow ab \neq 0$).
- **Associativity:** Multiplication is associative in $\mathbb{R}$.
- **Identity:** $1 \in \mathbb{R}^*$ satisfies $1 \cdot a = a$.
- **Inverse:** For every $a \in \mathbb{R}^*$, $\frac{1}{a} \in \mathbb{R}^*$ is the inverse ($a \cdot \frac{1}{a} = 1$).

■

Solved Problem 1.2.2.3: Compute the inverse of 3 in the group $\mathbb{Z}_{11}$ under multiplication.

Solution: We need x such that $3x \equiv 1 \pmod{11}$. Testing values: $3(1) = 3, 3(2) = 6, 3(3) = 9, 3(4) = 12$. Since $12 \equiv 1 \pmod{11}$, then $3(4) \equiv 1$. Thus, $3^{-1} = 4$.

■

1.2.3 Supplementary Problems

Supplementary Problem 1.2.3.1: Determine if the set of 2×2 matrices with integer entries forms a group under matrix addition.

Supplementary Problem 1.2.3.2: Does the set of integers $\mathbb{Z}$ form a group under subtraction?

1.3 Subgroups

1.3.1 Theory

Definition 1.3.1.1 (Subgroup): A subset H of a group G is called a **subgroup** if H itself forms a group under the same operation defined on G . We denote this as $H \leq G$.

Example 1.3.1.1: The set of even integers $2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$ is a subgroup of the integers $\mathbb{Z}$ under addition. It is closed under addition ($2m + 2n = 2(m + n)$) and inverses ($-2m = 2(-m)$).

Proposition 1.3.1.1 (Subgroup Test): A non-empty subset $H \subseteq G$ is a subgroup if and only if for all $a, b \in H$, $a \cdot b^{-1} \in H$.

Theorem 1.3.1.1 (Lagrange's Theorem): If G is a finite group and H is a subgroup of G , then the order (number of elements) of H divides the order of G . i.e., $|H|$ divides $|G|$.

Proof: The left cosets of H in G form a partition of G . The map $h \mapsto gh$ is a bijection between H and gH , so every coset has the same size $|H|$. If there are k distinct cosets, then $|G| = k |H|$. Thus $|H|$ divides $|G|$. ■

Example 1.3.1.2: Small Subgroup Attacks: Subgroups play a crucial role in preventing “small subgroup attacks” in cryptographic protocols. In Diffie-Hellman, if the group order has small factors, an attacker can confine the shared secret to a small subgroup and exhaustively search for it. We often choose groups of prime order or groups where the order has a large prime factor to avoid this.

1.3.2 Solved Problems

Solved Problem 1.3.2.1: Finding Subgroups Find all subgroups of $\mathbb{Z}_6$ (under addition).

Solution: The order of $\mathbb{Z}_6$ is 6. By Lagrange’s Theorem, possible orders for subgroups are divisors of 6: 1, 2, 3, 6.

- Order 1: $\{0\}$ (trivial subgroup).
- Order 2: Elements of order 2 satisfy $x + x = 2x = 0$. In $\mathbb{Z}_6$, $2(3) = 6 \equiv 0$. Subgroup: $\{0, 3\}$.
- Order 3: Elements of order 3 satisfy $3x = 0$. $2(3) = 6 \equiv 0$ and $4(3) = 12 \equiv 0$. Subgroup: $\{0, 2, 4\}$.
- Order 6: $\mathbb{Z}_6$ itself.

■

Solved Problem 1.3.2.2: Let $G = \mathbb{Z}_7^*$ (under multiplication). Find the subgroup generated by the element 2.

Solution: $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$. We calculate powers of 2 modulo 7: $2^1 = 2$, $2^2 = 4$, $2^3 = 8 \equiv 1 \pmod{7}$. Since we reached the identity, the subgroup generated by 2 is $\langle 2 \rangle = \{1, 2, 4\}$. Example order is 3.

■

Solved Problem 1.3.2.3: Propose a reason why we might want to use a subgroup of prime order for cryptography.

Solution: Prime order subgroups have no proper non-trivial subgroups (Lagrange’s Theorem: divisors of prime p are only 1 and p). This prevents small subgroup attacks where an attacker forces the computation into a smaller group to solve the discrete logarithm problem more easily. If the group has prime order, the attacker cannot force the key into a non-trivial subgroup.

■

1.3.3 Supplementary Problems

Supplementary Problem 1.3.3.1: List all subgroups of $\mathbb{Z}_{12}$.

Supplementary Problem 1.3.3.2: Show that the intersection of two subgroups H and K of a group G is also a subgroup of G .

1.4 Homomorphisms

1.4.1 Theory

Definition 1.4.1.1 (Homomorphism): A **homomorphism** is a structure-preserving map between two algebraic structures. Let $(G, \cdot)$ and $(H, *)$ be groups. A function $\varphi : G \rightarrow H$ is a group homomorphism if for all $a, b \in G$:

$$\varphi(a \cdot b) = \varphi(a) * \varphi(b)$$

Example 1.4.1.1: The map $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, \cdot)$ defined by $x \mapsto e^x$ is a homomorphism because $e^{x+y} = e^x \cdot e^y$.

Proposition 1.4.1.1 (Properties of Homomorphisms):

- $\varphi(e_G) = e_H$ (maps identity to identity)
- $\varphi(a^{-1}) = \varphi(a)^{-1}$ (maps inverses to inverses)

Proof:

1. $\varphi(e_G) = \varphi(e_G \cdot e_G) = \varphi(e_G) * \varphi(e_G)$. Multiplying by $\varphi(e_G)^{-1}$ gives $e_H = \varphi(e_G)$.
2. $e_H = \varphi(e_G) = \varphi(a \cdot a^{-1}) = \varphi(a) * \varphi(a^{-1})$. Thus $\varphi(a^{-1})$ is the inverse of $\varphi(a)$.

■

Kernel 1.4.1.2: The **Kernel** of φ , denoted $\ker(\varphi)$, is the set $\{g \in G \mid \varphi(g) = e_H\}$. It is always a normal subgroup of G .

Example 1.4.1.2: For the homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_2$ given by $\varphi(x) = x \bmod 2$, the kernel corresponds to the set of even integers $2\mathbb{Z}$, as these are exactly the integers mapped to 0.

Example 1.4.1.3: Homomorphic Encryption: Homomorphisms are implicit in many reduction steps in cryptography. For example, the security reduction of many protocols relies on mapping a complex problem instance to another structure where relations are preserved. They are also fundamental in constructing “Homomorphic Encryption” schemes, where computations can be performed on encrypted data ($\varphi(\text{enc}(m_1) \cdot \text{enc}(m_2)) = \varphi(\text{enc}(m_1 \cdot m_2))$), allowing operations on the ciphertext to map to operations on the plaintext.

1.4.2 Solved Problems

Solved Problem 1.4.2.1: Consider $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}_n$ defined by $\varphi(x) = x \bmod n$. Prove it is a homomorphism.

Solution: We need to check the homomorphism property $\varphi(x + y) = \varphi(x) +_n \varphi(y)$.

$$\varphi(x + y) = (x + y) \bmod n$$

From modular arithmetic properties: $(x + y) \bmod n = ((x \bmod n) + (y \bmod n)) \bmod n$.

This is exactly the definition of addition in $\mathbb{Z}_n$, i.e., $\varphi(x) +_n \varphi(y)$. Thus, φ is a homomorphism. ■

Solved Problem 1.4.2.2: Let $\varphi : \mathbb{R}^* \rightarrow \mathbb{R}^*$ be defined by $\varphi(x) = |x|$. Is φ a homomorphism under multiplication?

Solution: Check if $\varphi(xy) = \varphi(x)\varphi(y)$.

$$\text{LHS: } \varphi(xy) = |xy|.$$

$$\text{RHS: } \varphi(x)\varphi(y) = |x| \cdot |y|.$$

Since $|xy| = |x||y|$ for all real numbers, φ is a homomorphism. ■

Solved Problem 1.4.2.3: Find the kernel of the homomorphism $\varphi : \mathbb{Z}_{12} \rightarrow \mathbb{Z}_4$ defined by $\varphi(x) = x \bmod 4$.

Solution: We need to find all $x \in \mathbb{Z}_{12} = \{0, 1, \dots, 11\}$ such that $\varphi(x) = 0$ in $\mathbb{Z}_4$ (i.e., x is a multiple of 4).

Testing elements:

- $0 \bmod 4 = 0$ (in Kernel)
- $4 \bmod 4 = 0$ (in Kernel)
- $8 \bmod 4 = 0$ (in Kernel)

Any other element is not divisible by 4. Thus, $\ker(\varphi) = \{0, 4, 8\}$. ■

1.4.3 Supplementary Problems

Supplementary Problem 1.4.3.1: Determine if $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $\varphi(x) = 2x + 1$ is a group homomorphism.

Supplementary Problem 1.4.3.2: Let G be abelian and n be an integer. Show that $\varphi : G \rightarrow G$ defined by $\varphi(x) = x^n$ is a homomorphism.

1.5 Automorphisms

1.5.1 Theory

Definition 1.5.1.1 (Automorphism): An **automorphism** of a group G is an isomorphism from G to itself. That is, a bijection $\varphi : G \rightarrow G$ that is also a homomorphism.

Example 1.5.1.1: For the additive group of integers $\mathbb{Z}$, the map $\varphi(x) = -x$ is an automorphism. It is a bijection and preserves addition: $\varphi(x + y) = -(x + y) = (-x) + (-y) = \varphi(x) + \varphi(y)$.

Proposition 1.5.1.1: The set of all automorphisms of G , denoted $\text{Aut}(G)$, forms a group under function composition.

Proof:

1. **Closure:** The composition of two isomorphisms is an isomorphism.
2. **Associativity:** Function composition is associative.
3. **Identity:** The identity map $\text{id}(x) = x$ is an automorphism.
4. **Inverse:** The inverse function of an isomorphism is also an isomorphism.

■

Inner Automorphism 1.5.1.2: For any $g \in G$, the conjugation map $\varphi_{g(x)} = gxg^{-1}$ is an automorphism of G called an **inner automorphism**.

Example 1.5.1.2: In the symmetric group S_3 , conjugation by an element can permute the cycles. However, as noted in the solved problems, in any abelian group, the only inner automorphism is the identity map since $gxg^{-1} = xgg^{-1} = x$.

Example 1.5.1.3: Structure Weaknesses: Automorphisms describe the symmetries of the algebraic structure. In cryptography, understanding the automorphism group of the underlying structure (like an elliptic curve) is vital for understanding potential weaknesses or for optimizing arithmetic (e.g., using Frobenius automorphisms for faster scalar multiplication on Koblitz curves).

1.5.2 Solved Problems

Solved Problem 1.5.2.1: Let $G = \mathbb{Z}_5$ (additive). Find all automorphisms of $\mathbb{Z}_5$.

Solution: Any automorphism must map a generator to a generator. Generators of $\mathbb{Z}_5$ are $\{1, 2, 3, 4\}$ (since 5 is prime). Maps $\varphi(x) = kx$ where k is a generator:

- $\sigma_1(x) = 1x = x$ (Identity)
- $\sigma_2(x) = 2x \bmod 5$
- $\sigma_3(x) = 3x \bmod 5$

- $\sigma_4(x) = 4x \bmod 5$

Each of these is an automorphism. Thus $|\text{Aut}(\mathbb{Z}_5)| = 4$. ■

Solved Problem 1.5.2.2: Determine all automorphisms of the group $\mathbb{Z}_6$.

Solution: Automorphisms of $\mathbb{Z}_n$ are determined by where they map 1. Since 1 generates $\mathbb{Z}_n$, $\varphi(1)$ must be a generator of $\mathbb{Z}_n$. The generators of $\mathbb{Z}_6$ are elements coprime to 6: $\{1, 5\}$. Two possible maps:

- $\sigma_1(x) = 1 \cdot x = x$ (Identity)
- $\sigma_5(x) = 5 \cdot x \equiv -x \bmod 6$.

Thus, $\text{Aut}(\mathbb{Z}_6) \simeq \mathbb{Z}_2$. ■

Solved Problem 1.5.2.3: Show that for an abelian group G , the only inner automorphism is the identity map.

Solution: Let G be abelian. An inner automorphism is defined by $\varphi_{g(x)} = gxg^{-1}$ for some $g \in G$. Since G is abelian, $gx = xg$. Therefore, $\varphi_{g(x)} = (xg)g^{-1} = x(gg^{-1}) = xe = x$. This is the identity map for all $g \in G$. ■

1.5.3 Supplementary Problems

Supplementary Problem 1.5.3.1: Prove that the map $\varphi : C \rightarrow C$ given by $\varphi(z) = \bar{z}$ (complex conjugation) is an automorphism of the additive group of complex numbers.

Supplementary Problem 1.5.3.2: Let $G = S_3$. Find the inner automorphism defined by $g = (12)$.

1.6 Cayley's Theorem

1.6.1 Theory

Theorem 1.6.1.1 (Cayley's Theorem): Every group G is isomorphic to a subgroup of the symmetric group acting on G . In other words, every group can be considered as a group of permutations.

Example 1.6.1.1: For the cyclic group $\mathbb{Z}_3 = \{0, 1, 2\}$, Cayley's theorem tells us we can view its elements as permutations of the set $\{0, 1, 2\}$. For instance, adding 1 corresponds to the cycle $(0\ 1\ 2)$.

Proposition 1.6.1.1 (Permutation Representation): Specifically, for each $g \in G$, we can define a permutation $\lambda_g : G \rightarrow G$ by $\lambda_{g(x)} = g \cdot x$. The mapping $g \mapsto \lambda_g$ is an injective homomorphism from G into S_G (the group of all permutations of the set G).

Proof:

1. **Well-defined:** λ_g is a permutation because it has an inverse $\lambda_{g^{-1}}$.
2. **Homomorphism:** $\lambda_{gh}(x) = (gh)x = g(hx) = \lambda_{g(\lambda_h(x))}$. Thus $g \mapsto \lambda_g$ preserves operations.
3. **Injective:** If $\lambda_g = \lambda_h$, then $\lambda_{g(e)} = \lambda_{h(e)} \Rightarrow g = h$.

Thus G is isomorphic to the image $\{\lambda_g \mid g \in G\}$, which is a subgroup of S_G . ■

Example 1.6.1.2: Significance: Cayley's theorem is theoretically significant as it allows us to represent abstract groups concretely using permutations. In computational group theory, permutation representations are often easier to store and manipulate in a computer than abstract definitions, allowing for efficient algorithms to determine group properties.

1.6.2 Solved Problems

Solved Problem 1.6.2.1: Consider the group $G = \{1, -1\}$ under multiplication. Find its permutation representation.

Solution: Elements: $1, -1$. Permutations correspond to left multiplication:

- $\lambda_1: 1 \rightarrow 1, -1 \rightarrow -1$. Disjoint cycle form: $(1)(-1)$ or e .
- $\lambda_{-1}: 1 \rightarrow -1, -1 \rightarrow 1$. Disjoint cycle form: $(1, -1)$.

Thus $G \simeq \{e, (1, -1)\} \subset S_2$. ■

Solved Problem 1.6.2.2: Write down the permutation representation of $\mathbb{Z}_3 = \{0, 1, 2\}$ given by Cayley's Theorem.

Solution: Elements: 0, 1, 2. Operation is addition mod 3.

- $\lambda_0: 0 \rightarrow 0, 1 \rightarrow 1, 2 \rightarrow 2 \rightarrow (0)(1)(2) = e$
- $\lambda_1: 0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 0 \rightarrow (0\ 1\ 2)$
- $\lambda_2: 0 \rightarrow 2, 1 \rightarrow 0, 2 \rightarrow 1 \rightarrow (0\ 2\ 1)$

■

Solved Problem 1.6.2.3: Explain why the image of the map $g \mapsto \lambda_g$ is a subgroup of S_G .

Solution: The map $\varphi : G \rightarrow S_G$ given by $\varphi(g) = \lambda_g$ is a homomorphism. A fundamental property of homomorphisms is that the image of a group is always a subgroup of the codomain. Since G is a group and φ is a homomorphism, $\text{Im}(\varphi)$ is a subgroup of S_G .

■

1.6.3 Supplementary Problems

Supplementary Problem 1.6.3.1: Let $G = \{e, a, b, c\}$ be the Klein 4-group where $x^2 = e$ for all x . Construct the left regular representation (Cayley representation) for a .

Supplementary Problem 1.6.3.2: True or False: If G has order n , then G is isomorphic to a subgroup of S_n .

1.7 Permutation Groups

1.7.1 Theory

Definition 1.7.1.1 (Permutation Group): A **permutation group** is a group G whose elements are permutations of a given set M and whose group operation is function composition. The symmetric group S_n consists of all permutations of a set of n elements and has order $n!$.

Example 1.7.1.1: The group S_3 contains $3! = 6$ elements. These are the Identity e , three 2-cycles $(1\ 2), (1\ 3), (2\ 3)$, and two 3-cycles $(1\ 2\ 3), (1\ 3\ 2)$.

Proposition 1.7.1.1 (Cycle Decomposition): Any permutation can be decomposed into disjoint cycles. The order of a permutation is the least common multiple (LCM) of the lengths of these disjoint cycles.

Proof: Consider the limit of the sequence $x, \sigma(x), \sigma^2(x), \dots$ for any x . Since the set is finite, it must loop back to x . This forms a cycle. Removing these elements and repeating the process decomposes the entire permutation into disjoint cycles. For disjoint cycles, operations commute. Order is the smallest k such that $\text{LCM}(l_1, \dots, l_m)$ divides k . ■

Example 1.7.1.2: Ciphers: Permutation groups are the basis of many classical ciphers (like the Enigma machine, which used complex permutations). In modern design, substitution-permutation networks (SPNs) used in block ciphers like AES rely heavily on mixing (permutations) and substitution (S-boxes, which are often defined via other algebraic properties) to achieve confusion and diffusion.

1.7.2 Solved Problems

Solved Problem 1.7.2.1: In S_5 , let $\sigma = (1\ 2\ 3)(4\ 5)$. Find the order of σ and compute σ^2 .

Solution: This is a composition of a 3-cycle and a 2-cycle, which are disjoint. Order of $\sigma = \text{lcm}(3, 2) = 6$.

Computing σ^2 : $\sigma^2 = ((1\ 2\ 3)(4\ 5))^2$ Since they are disjoint, they commute: $= (1\ 2\ 3)^2(4\ 5)^2 = (1\ 3\ 2)(e) = (1\ 3\ 2)$ ■

Solved Problem 1.7.2.2: Calculate the product of permutations $\pi = (1\ 3\ 4)$ and $\sigma = (1\ 2)(3\ 4)$ in S_4 .

Solution: Compute from right to left (standard function composition): $\pi\sigma$.

- $1 \xrightarrow{\sigma} 2 \xrightarrow{\pi} 2$

- $2 \xrightarrow{\sigma} 1 \xrightarrow{\pi} 3$
- $3 \xrightarrow{\sigma} 4 \xrightarrow{\pi} 1$
- $4 \xrightarrow{\sigma} 3 \xrightarrow{\pi} 4$

Result: $(1\ 2\ 3)(4) = (1\ 2\ 3)$. ■

Solved Problem 1.7.2.3: Find the order of the permutation $(1\ 2\ 4)(3\ 5\ 6)$ in S_6 .

Solution: The permutation is a product of two disjoint cycles of length 3. Order = $\text{lcm}(3, 3) = 3$. ■

Solved Problem 1.7.2.4: How many elements does S_4 have? List 5 different elements.

Solution: $|S_4| = 4! = 4 \times 3 \times 2 \times 1 = 24$.

Examples:

1. e (Identity)
2. $(1\ 2)$
3. $(1\ 3)$
4. $(1\ 2\ 3)$
5. $(1\ 2)(3\ 4)$

■

1.7.3 Supplementary Problems

Supplementary Problem 1.7.3.1: Express the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 1 & 2 \end{pmatrix}$ as a product of disjoint cycles.

Supplementary Problem 1.7.3.2: Find the inverse of the permutation $(1\ 2\ 3)(4\ 5)$.

2 Ring Theory

2.1 Introduction

Ring theory studies algebraic structures with two binary operations: addition and multiplication. Rings generalize the arithmetic of integers. They are crucial in cryptography because most cryptographic algorithms (like RSA, AES, ECC) operate over specific finite rings or fields (which are special types of rings).

2.2 Definition and Examples of Rings

2.2.1 Theory

Ring 2.2.1.1: A **ring** is a set R equipped with two binary operations, usually denoted by $+$ (addition) and $\cdot$ (multiplication), satisfying:

1. $(R, +)$ is an abelian group (additive identity is 0 , inverse of a is $-a$).
2. $(R, \cdot)$ satisfies associativity $(a \cdot (b \cdot c) = (a \cdot b) \cdot c)$.
3. Distributive Laws hold: $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

Example 2.2.1.1: The set of integers $\mathbb{Z}$ with standard addition and multiplication is a ring. The set of even integers $2\mathbb{Z}$ is also a ring (without unity). The set of $n \times n$ matrices $M_{n(\mathbb{R})}$ is a non-commutative ring.

Definition 2.2.1.2:

- If there is a multiplicative identity (denoted 1 such that $1 \cdot a = a \cdot 1 = a$), it is a **ring with unity**.
- If multiplication is commutative, it is a **commutative ring**.

Example 2.2.1.2: The ring of integers $\mathbb{Z}$ is a commutative ring with unity 1 . The ring of even integers $2\mathbb{Z}$ is a commutative ring but has no multiplicative identity. The ring of 2×2 matrices $M_2(\mathbb{R})$ has a unity (the identity matrix I) but is not commutative.

Definition 2.2.1.3 (Field): A **field** is a commutative ring with unity where every non-zero element has a multiplicative inverse.

Example 2.2.1.3:

- **Finite Fields:** $\text{GF}(p)$ (or $\mathbb{Z}_p$) where p is prime, are fields widely used in ECC.
- **Integers modulo n :** $\mathbb{Z}_n$ is a ring used in RSA.

2.2.2 Solved Problems

Solved Problem 2.2.2.1: In the ring $\mathbb{Z}_{12}$, compute 5×7 and explain why $\mathbb{Z}_{12}$ is not a field.

Solution: Arithmetic is done modulo 12:

$$5 \times 7 = 35$$

$$35 = 2(12) + 11 \equiv 11 \pmod{12}$$

.

$\mathbb{Z}_{12}$ is not a field because not every non-zero element has a multiplicative inverse. For example, consider 2. There is no integer x such that $2x \equiv 1 \pmod{12}$ because $2x$ is always even, while $1 + 12k$ is always odd (or simply because $\gcd(2, 12) \neq 1$). ■

Solved Problem 2.2.2.2: Consider the set $R = \mathbb{Z} \times \mathbb{Z}$ with component-wise addition and multiplication. Is R a ring?

Solution: Yes.

- Addition $(a, b) + (c, d) = (a + c, b + d)$ forms an abelian group (inherited from $\mathbb{Z}$).
- Multiplication $(a, b) \cdot (c, d) = (ac, bd)$ is associative.
- Distributivity holds:

$$(a, b) \cdot ((c, d) + (e, f)) = (a, b) \cdot (c + e, d + f) = (ac + ae, bd + bf)$$

$$(a, b)(c, d) + (a, b)(e, f) = (ac, bd) + (ae, bf) = (ac + ae, bd + bf)$$

Thus, it is a ring. ■

Solved Problem 2.2.2.3: Find all units (invertible elements) in the ring $\mathbb{Z}_{10}$.

Solution: An element $a \in \mathbb{Z}_n$ is a unit if and only if $\gcd(a, n) = 1$. For $n = 10$:

- $\gcd(1, 10) = 1$ (Unit)

- $\gcd(2, 10) = 2$
- $\gcd(3, 10) = 1$ (Unit)
- $\gcd(4, 10) = 2$
- $\gcd(5, 10) = 5$
- $\gcd(6, 10) = 2$
- $\gcd(7, 10) = 1$ (Unit)
- $\gcd(8, 10) = 2$
- $\gcd(9, 10) = 1$ (Unit)

The group of units is $\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$. ■

2.2.3 Supplementary Problems

Supplementary Problem 2.2.3.1: Determine if the set of even integers $2\mathbb{Z}$ is a ring under standard addition and multiplication. Is it a ring with unity?

Supplementary Problem 2.2.3.2: Solve the equation $x^2 - 5x + 6 = 0$ in the ring $\mathbb{Z}_{\{12\}}$. (Hint: There may be more than 2 solutions).

2.3 Ideals and Quotient Rings

2.3.1 Theory

Definition 2.3.1.1 (Ideal): An **ideal** I of a ring R is a subring such that for all $r \in R$ and $x \in I$, both $r \cdot x$ and $x \cdot r$ are in I . Ideals “absorb” multiplication by ring elements.

Example 2.3.1.1: The set of multiples of n , denoted $n\mathbb{Z}$, is an ideal of $\mathbb{Z}$. For any integer r and any multiple nk , the product $r(nk) = n(rk)$ is still a multiple of n .

Definition 2.3.1.2 (Quotient Ring): Given an ideal I , we can form the **quotient ring** $\frac{R}{I}$. The elements of $\frac{R}{I}$ are cosets $a + I$, and operations are defined naturally:

- $(a + I) + (b + I) = (a + b) + I$
- $(a + I) \cdot (b + I) = (a \cdot b) + I$

Example 2.3.1.2: If $R = \mathbb{Z}$ and $I = 3\mathbb{Z}$, the quotient ring $\frac{\mathbb{Z}}{3}\mathbb{Z}$ has elements $\{0 + 3\mathbb{Z}, 1 + 3\mathbb{Z}, 2 + 3\mathbb{Z}\}$. This is isomorphic to the integers modulo 3, $\mathbb{Z}_3$.

Proposition 2.3.1.1 (Kernel is an Ideal): The kernel of any ring homomorphism $\varphi : R \rightarrow S$ is an ideal of R .

Proof: Let $K = \ker(\varphi)$.

1. **Subgroup:** K is an additive subgroup (standard group theory result).
2. **Absorption:** Let $r \in R, k \in K$. $\varphi(r \cdot k) = \varphi(r) \cdot \varphi(k) = \varphi(r) \cdot 0 = 0$. $\varphi(k \cdot r) = \varphi(k) \cdot \varphi(r) = 0 \cdot \varphi(r) = 0$. Thus $r \cdot k \in K$ and $k \cdot r \in K$, so K is an ideal. ■

Example 2.3.1.3: AES S-Box: Quotient rings are fundamental in constructing finite fields consisting of polynomial bases, e.g., $\text{GF}(2^8) \simeq \mathbb{Z}_2 \frac{[x]}{P(x)}$ where $P(x)$ is an irreducible polynomial. This structure is the basis for the AES S-box.

2.3.2 Solved Problems

Solved Problem 2.3.2.1: Show that the set of even integers $2\mathbb{Z}$ is an ideal of $\mathbb{Z}$.

Solution:

1. Subring Check:

- Non-empty: $0 \in 2\mathbb{Z}$.
- Closed under subtraction: If $2a, 2b \in 2\mathbb{Z}$, then $2a - 2b = 2(a - b) \in 2\mathbb{Z}$.

2. Absorption Check:

- Let $x \in 2\mathbb{Z}$ (so $x = 2k$) and $r \in \mathbb{Z}$.
- $rx = r(2k) = 2(rk) \in 2\mathbb{Z}$.

Since it satisfies both conditions (and multiplication is commutative), $2\mathbb{Z}$ is an ideal. ■

Solved Problem 2.3.2.2: Consider $R = \mathbb{Z}$ and $I = 5\mathbb{Z}$. Describe the quotient ring $\mathbb{Z}/I$.

Solution: The elements of $\mathbb{Z}/5\mathbb{Z}$ are the cosets $a + 5\mathbb{Z}$. Two cosets are equal if their representatives differ by a multiple of 5. Distinct cosets correspond to the remainders modulo 5: $0 + 5\mathbb{Z}, 1 + 5\mathbb{Z}, 2 + 5\mathbb{Z}, 3 + 5\mathbb{Z}, 4 + 5\mathbb{Z}$.

Addition and multiplication exactly mirror arithmetic in $\mathbb{Z}_5$. Thus, $\frac{\mathbb{Z}}{5}\mathbb{Z} \simeq \mathbb{Z}_5$. ■

Solved Problem 2.3.2.3: In $\mathbb{Z}[x]$, let $I = \langle x^2 + 1 \rangle$. Describe the elements of $\frac{\mathbb{Z}[x]}{I}$.

Solution: Elements of the quotient are polynomials modulo $x^2 + 1$. By the Division Algorithm for polynomials, any $f(x) \in \mathbb{Z}[x]$ can be written as: $f(x) = q(x)(x^2 + 1) + r(x)$, where $\deg(r) < 2$.

Thus, every coset has a unique representative of the form $ax + b$ where $a, b \in \mathbb{Z}$. Multiplication satisfies $x^2 \equiv -1$. This ring is isomorphic to the Gaussian integers $\mathbb{Z}[i]$ via the map $ax + b \mapsto bi + a$. ■

2.3.3 Supplementary Problems

Supplementary Problem 2.3.3.1: Let R be a commutative ring with unity. Show that R is a field if and only if its only ideals are $\{0\}$ and R itself.

Supplementary Problem 2.3.3.2: Let I be the set of matrices in $M_2(\mathbb{R})$ with the first column equal to zero. Is I a left ideal, right ideal, or two-sided ideal?

2.4 Euclidean Rings

2.4.1 Theory

Euclidean Domain 2.4.1.1: A **Euclidean Domain** (or Euclidean Ring) is an integral domain R equipped with a Euclidean value function (or norm) $N : R \setminus \{0\} \rightarrow \mathbb{N}$ allowing a division algorithm. For any $a, b \in R$ with $b \neq 0$, there exist $q, r \in R$ such that:

$$a = bq + r$$

where either $r = 0$ or $N(r) < N(b)$.

Example 2.4.1.1: The ring of integers $\mathbb{Z}$ is a Euclidean domain with norm $N(a) = |a|$. The ring of polynomials $F[x]$ over a field F is a Euclidean domain with norm $N(f) = \deg(f)$.

Theorem 2.4.1.1 (Extended Euclidean Algorithm (EEA)): The Extended Euclidean Algorithm (EEA) finds the greatest common divisor (GCD) and coefficients x, y such that $ax + by = \gcd(a, b)$.

Proof: The algorithm proceeds by successive divisions: $r_{i-2} = q_i r_{i-1} + r_i$. Since $N(r_i)$ is a strictly decreasing sequence of non-negative integers, it must terminate with $r_k = 0$. The last non-zero remainder r_{k-1} divides all previous remainders and linear combinations, so it is the GCD. Back-substitution yields x, y . ■

Example 2.4.1.2: Key Generation:

- **RSA Key Generation:** The EEA is used to compute the modular multiplicative inverse of the public exponent e to find the private key d ($ed \equiv 1 \pmod{\varphi(n)}$).
- **Determining Inverses:** In any finite field, finding inverses is done via the Euclidean algorithm.

2.4.2 Solved Problems

Solved Problem 2.4.2.1: Show that $\mathbb{Z}$ is a Euclidean domain.

Solution: Let $N(a) = |a|$. The standard division algorithm for integers states that for any $a, b \in \mathbb{Z} (b \neq 0)$, there exist q, r such that $a = bq + r$ with $0 \leq r < |b|$. Thus either $r = 0$ or $N(r) = |r| < |b| = N(b)$, satisfying the Euclidean condition. ■

Solved Problem 2.4.2.2: Use the Extended Euclidean Algorithm to find the inverse of 17 modulo 26.

Solution: We seek x such that $17x \equiv 1 \pmod{26}$. This is equivalent to finding x, y such that $17x + 26y = 1$. Steps:

1. $26 = 1(17) + 9$

$$2. 17 = 1(9) + 8$$

$$3. 9 = 1(8) + 1 \text{ (GCD is 1)}$$

Back substitution: $1 = 9 - 8$ $1 = 9 - (17 - 9) = 2(9) - 17$ $1 = 2(26 - 17) - 17 = 2(26) - 3(17)$

Thus $17(-3) + 26(2) = 1$. Modulo 26: $17(-3) \equiv 1$. Inverse is $-3 \equiv 23 \pmod{26}$. ■

Solved Problem 2.4.2.3: Prove that the ring of Gaussian Integers $\mathbb{Z}[i]$ is a Euclidean domain.

Solution: Define norm $N(a + bi) = a^2 + b^2$. For any $z, w \in \mathbb{Z}[i]$ ($w \neq 0$), consider the complex number $\frac{z}{w}$. This point lies somewhere in the complex plane. The Gaussian integers form a square lattice in $\mathbb{C}$. Any point in the plane is at distance at most $\frac{\sqrt{2}}{2}$ from the nearest lattice point. Let q be the Gaussian integer closest to $\frac{z}{w}$. Then $|\frac{z}{w} - q| \leq \frac{\sqrt{2}}{2}$. Let $r = z - wq$. Then $|r| = |w| |\frac{z}{w} - q| \leq |w| \frac{\sqrt{2}}{2}$. Since $\frac{\sqrt{2}}{2} < 1$, $|r| < |w|$, so $N(r) < N(w)$. ■

2.4.3 Supplementary Problems

Supplementary Problem 2.4.3.1: Using the Euclidean Algorithm, find $\gcd(123, 456)$.

Supplementary Problem 2.4.3.2: In $\mathbb{Z}[i]$, divide $11 + 3i$ by $1 + i$ to find a quotient and remainder.

2.5 Polynomial Rings

2.5.1 Theory

Definition 2.5.1.1 (Polynomial Ring): A **Polynomial Ring** $R[x]$ consists of polynomials with coefficients in a ring R . The division algorithm for polynomials holds if R is a field.

Example 2.5.1.1: $\mathbb{Z}[x]$ is the ring of polynomials with integer coefficients. $\mathbb{R}[x]$ is the ring of polynomials with real coefficients.

Definition 2.5.1.2 (Irreducible Polynomial): A polynomial $P(x)$ is **irreducible** over a field F if it cannot be factored into two non-constant polynomials in $F[x]$.

Example 2.5.1.2: $x^2 + 1$ is irreducible over the reals $\mathbb{R}$ because it has no real roots. However, it is reducible over the complex numbers $\mathbb{C}$ as $(x - i)(x + i)$ and over $\mathbb{Z}_2$ as $(x + 1)^2 = x^2 + 1$.

Theorem 2.5.1.1 (Finite Field Construction): Finite fields of order p^n , denoted $\text{GF}(p^n)$, are constructed as $\frac{\mathbb{Z}_p[x]}{\langle P(x) \rangle}$ where $P(x)$ is an irreducible polynomial of degree n over $\mathbb{Z}_p$.

Proof: Let $F = \frac{\mathbb{Z}_p[x]}{\langle P(x) \rangle}$. F is a commutative ring with unity. Let $f(x) \in F$ be non-zero (meaning it is not a multiple of $P(x)$). Since $P(x)$ is irreducible and $\deg(f) < \deg(P)$, $\gcd(f, P) = 1$. By EEA for polynomials, there exist $a(x), b(x)$ such that $a(x)f(x) + b(x)P(x) = 1$. Taking this equation modulo $P(x)$, we get $a(x)f(x) \equiv 1$. Thus every non-zero element has an inverse, making F a field. ■

Example 2.5.1.3: Cryptography:

- **AES:** Operates in $\text{GF}(2^8)$ constructed with the irreducible polynomial $x^8 + x^4 + x^3 + x + 1$.
- **ECC:** Uses fields $\text{GF}(2^m)$ or $\text{GF}(p)$.

2.5.2 Solved Problems

Solved Problem 2.5.2.1: In $\mathbb{Z}_2[x]$, perform addition and multiplication of $f(x) = x^2 + x + 1$ and $g(x) = x + 1$.

Solution: Coefficients are in $\mathbb{Z}_2 = \{0, 1\}$.

Addition: $f(x) + g(x) = (x^2 + x + 1) + (x + 1) = x^2 + 2x + 2$. Since $2 \equiv 0 \pmod{2}$, $f(x) + g(x) = x^2$.

Multiplication: $f(x) \cdot g(x) = (x^2 + x + 1)(x + 1) = x^3 + x^2 + x + x^2 + x + 1 = x^3 + 2x^2 + 2x + 1 = x^3 + 1$. ■

Solved Problem 2.5.2.2: Factor $x^2 - 1$ in $\mathbb{Z}_5[x]$.

Solution: We use the difference of squares formula: $x^2 - 1 = (x - 1)(x + 1)$. In $\mathbb{Z}_5$, $-1 \equiv 4$. So $x^2 - 1 = (x + 4)(x + 1)$. ■

Solved Problem 2.5.2.3: Determine if $x^3 + x + 1$ is irreducible in $\mathbb{Z}_2[x]$.

Solution: A polynomial of degree 2 or 3 is reducible over a field if and only if it has a root in that field. Test elements of $\mathbb{Z}_2 = \{0, 1\}$:

- $x = 0 : 0^3 + 0 + 1 = 1 \neq 0$.
- $x = 1 : 1^3 + 1 + 1 = 3 \equiv 1 \neq 0$.

Since there are no roots, the polynomial is irreducible. ■

2.5.3 Supplementary Problems

Supplementary Problem 2.5.3.1: List all irreducible polynomials of degree 2 in $\mathbb{Z}_2[x]$.

Supplementary Problem 2.5.3.2: Construct the multiplication table for the field $\text{GF}(4)$ constructed as $\mathbb{Z}_2[\alpha]/\langle \alpha^2 + \alpha + 1 \rangle$.